



Examples in Context

Let's explore how these concepts work together in a practical scenario. Imagine you're working with a simple online bookstore database where you need to both set up the structure and manage the data.

Learning Note

We're using full-length SQL commands to clearly demonstrate the differences between command categories. In real development environments, you might use shorter syntax, command-line tools, or graphical interfaces, but understanding the complete syntax helps you grasp what's actually happening.

Database Level Operations (Structure)

First, you would define your database structure using DDL commands:

```
1  -- Creating a new table (DDL - Database Level)
2  CREATE TABLE books (
3      book_id SERIAL PRIMARY KEY,
4      title VARCHAR(100) NOT NULL,
5      author VARCHAR(100) NOT NULL,
6      price DECIMAL(10,2),
7      category VARCHAR(50),
8      publication_date DATE
9  );
10
11 -- Modifying table structure by adding a column (DDL - Database Level)
12 ALTER TABLE books ADD COLUMN in_stock BOOLEAN DEFAULT true;
13
14 -- Setting up access permissions (DCL - Database Level)
15 GRANT SELECT ON books TO store_staff;
16 GRANT INSERT, UPDATE, DELETE ON books TO store_managers;
```

Data Level Operations (Content)

After defining the structure, you would use DML commands to manipulate the actual data, implementing all four CRUD operations:

```
1  -- Adding a new book record (DML - Data Level - CREATE)
2  INSERT INTO books (title, author, price, category, publication_date)
3  VALUES ('Database Fundamentals', 'Jane Smith', 29.99, 'Technology',
4
5  -- Retrieving book information (DQL - Data Level - READ)
6  SELECT title, author, price FROM books WHERE category = 'Technology'
7
8  -- Updating a book's price (DML - Data Level - UPDATE)
9  UPDATE books SET price = 24.99 WHERE title = 'Database Fundamentals'
10
11 -- Removing a book record (DML - Data Level - DELETE)
12 DELETE FROM books WHERE title = 'Database Fundamentals';
13
14 -- Managing a transaction for multiple operations (TCL - Data Level)
15 BEGIN;
16 INSERT INTO books (title, author, price, category, publication_date)
17 VALUES ('SQL Essentials', 'John Doe', 34.99, 'Technology', '2024-01-01');
18 UPDATE books SET price = price * 0.9 WHERE category = 'Technology';
19 COMMIT;
```